STATE INTERPRETER

State Interpreter for Interpretable Machine Condition Analysis

A reproducible PHM pipeline for representation learning and state validation

Yuxin Song

M.Sc. Autonomous Systems · University of Stuttgart

Python | PyTorch | Signal Processing | Model Validation

WORK SAMPLE

Vibration
↓
STFT
↓
$z \in \mathbb{R}^8$
↓
Relative State

From raw vibration to a state that can be reasoned about

Raw vibration is high-dimensional, operating-condition dependent and not directly actionable.

1 REPRESENT

Compress two-channel vibration into a compact latent vector without using lifetime as a training target.

2 INTERPRET

Convert the latent trajectory into relative Level, Trend and Movement signals that remain causal and traceable.

3 VALIDATE

Test stability across random seeds, unseen bearings and operating conditions while preventing leakage.

Research question

Can a learned embedding support a continuous, interpretable and reproducible relative machine state?

Real run-to-failure data with leakage-controlled preprocessing

XJTU-SY
2-channel vibration
25.6 kHz

→

Recording
1.28 s · 32,768
samples/channel

→

Log-STFT
Hann · FFT 1024
hop 512

→

Pooling
fixed tensor
[2, 32, 32]

→

Normalization
fit on training
bearings only

DATA COVERAGE

3 operating conditions
5 bearings per condition
One recording per minute
Run-to-failure ordering

SPLIT DISCIPLINE

Train / validation / test bearings are disjoint. Test bearings do not contribute normalization or checkpoint selection.

CAUSALITY

The interpreter uses only current and past embeddings. Each new bearing is reset and calibrated independently.

Normalized lifetime is used only as an evaluation proxy—not as a physical damage label.

Sources: XJTU-SY adapter, STFT module and experiment configs

SYSTEM ARCHITECTURE

A modular pipeline separates representation from interpretation



COMPLETED CORE

Deterministic preprocessing
Small convolutional autoencoder
Relative state baseline
Immutable experiment records

EVALUATED ALTERNATIVES

Left-to-right Gaussian HMM
Causal predictive GRU
Signed degradation axis
Residual-GRU surprise

DESIGN BOUNDARY

The encoder learns representation. The interpreter assigns relative temporal meaning. Either module can be replaced independently.

training target: STFT reconstruction

evaluation proxy: normalized lifetime

output: relative—not absolute—health

Sources: autoencoder.py · relative_temporal.py · hmm_temporal.py · gru_temporal.py

REPRESENTATION LEARNING

The z=8 embedding carries ordering information—but not a universal health scale

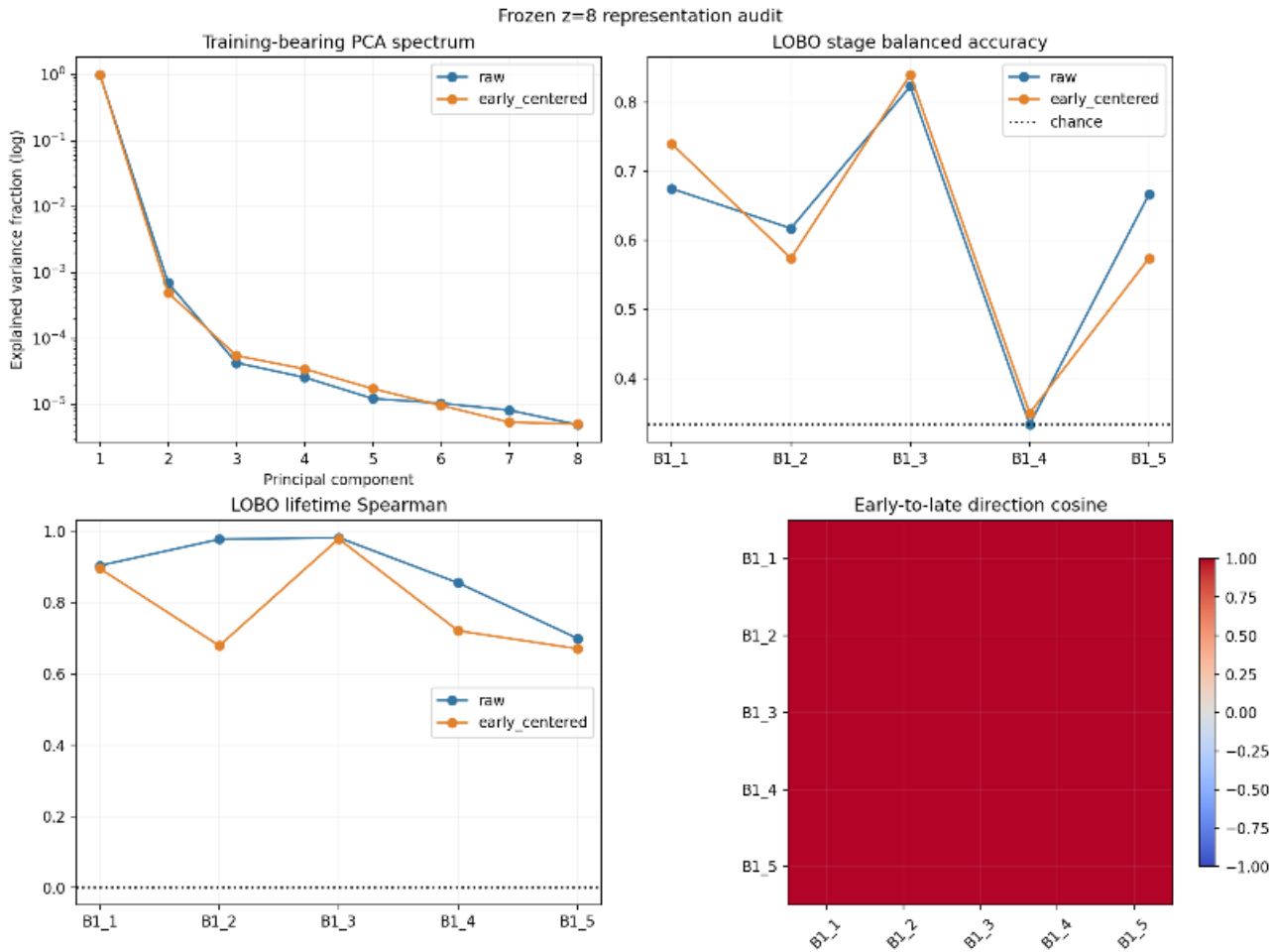


Figure: 616 frozen latent observations; no retraining during audit.

OBSERVATION

The first principal component dominates: the frozen representation is effectively close to one-dimensional.

USEFUL SIGNAL

Leave-one-bearing-out lifetime ordering remains positive across all five bearings in the audited condition.

LIMIT

Absolute stage classification is weak for Bearing1_4. Shared direction does not guarantee shared calibration.

Source: results/2026-09-17_z8_representation_audit

Interpreter comparison: simple assumptions first, temporal models second

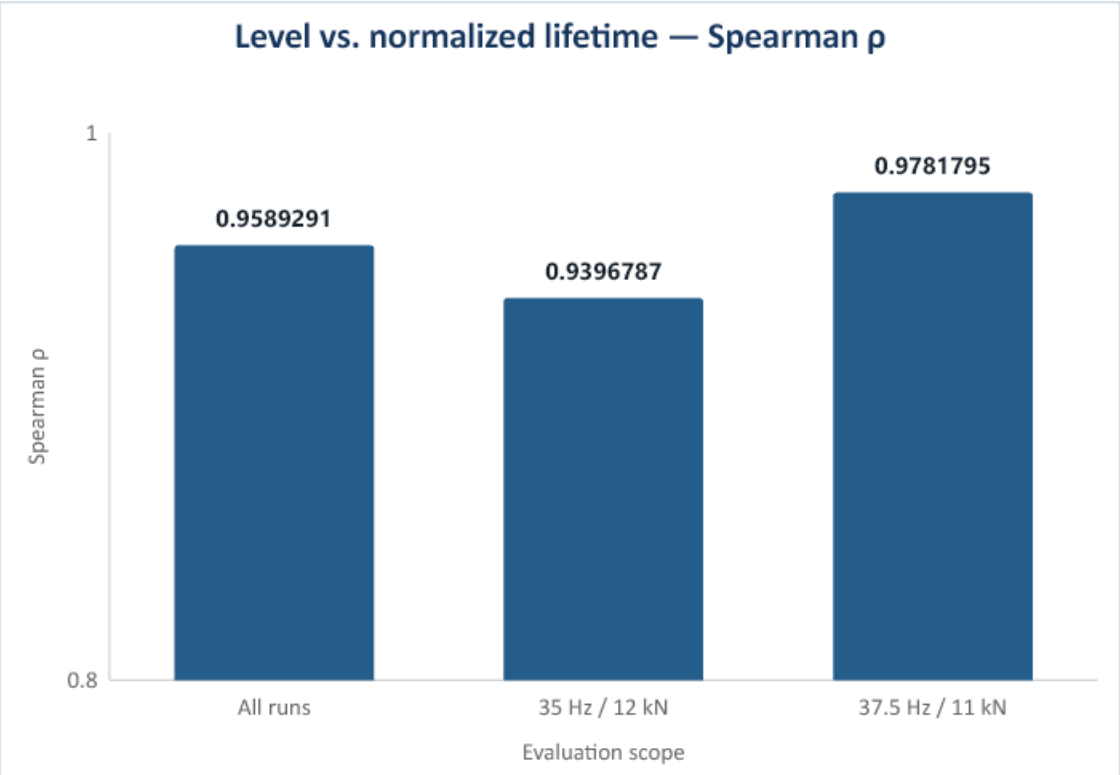
Method	Input → output	Assumption	Observed outcome
Distance baseline	z sequence → distance, slope, step norm	Early samples define a bearing-specific reference	Strong relative ordering; transparent and causal
3-state HMM	distance features → filtered stage probabilities	Monotonic stage transitions with Gaussian emissions	Smoother, but overconfident and one collapse case
Predictive GRU	past z → next-z forecast / hidden state	Dynamics improve state estimation	Did not consistently outperform the simple baseline
Relative State v2	signed axis + causal trend + residual surprise	Direction and deviation should be separated	Integrated relative state; semantics remain relative

Selection rule: keep added complexity only when it beats an explicit simple baseline on unseen bearings.

Sources: interpreter implementations and exploratory reports

EXPERIMENTAL DESIGN

Validation is designed around unseen bearings, seeds and conditions



30 TRAININGS

2 conditions × 5 LOBO folds × 3 random seeds
Each fold: 3 train · 1 validation · 1 test bearing

FAIR COMPARISON

Fixed z=8 architecture, 5 epochs and interpreter parameters (15 calibration steps / 10-step window).

TRACEABILITY

Configuration, execution record and result directories are separated. Blind-test data is protected by explicit gates and hashes.

95% CI for all 30 runs: [0.941, 0.977] · runs within a fold reuse data and are not independent samples.

Source: 2026-09-12 v0.1 stability study

A preregistered blind evaluation confirms strong relative ordering

BLIND TEST

Bearing2_5 · 339 measurements
15 calibration + 324 READY states

ARM A · FROZEN SOURCE

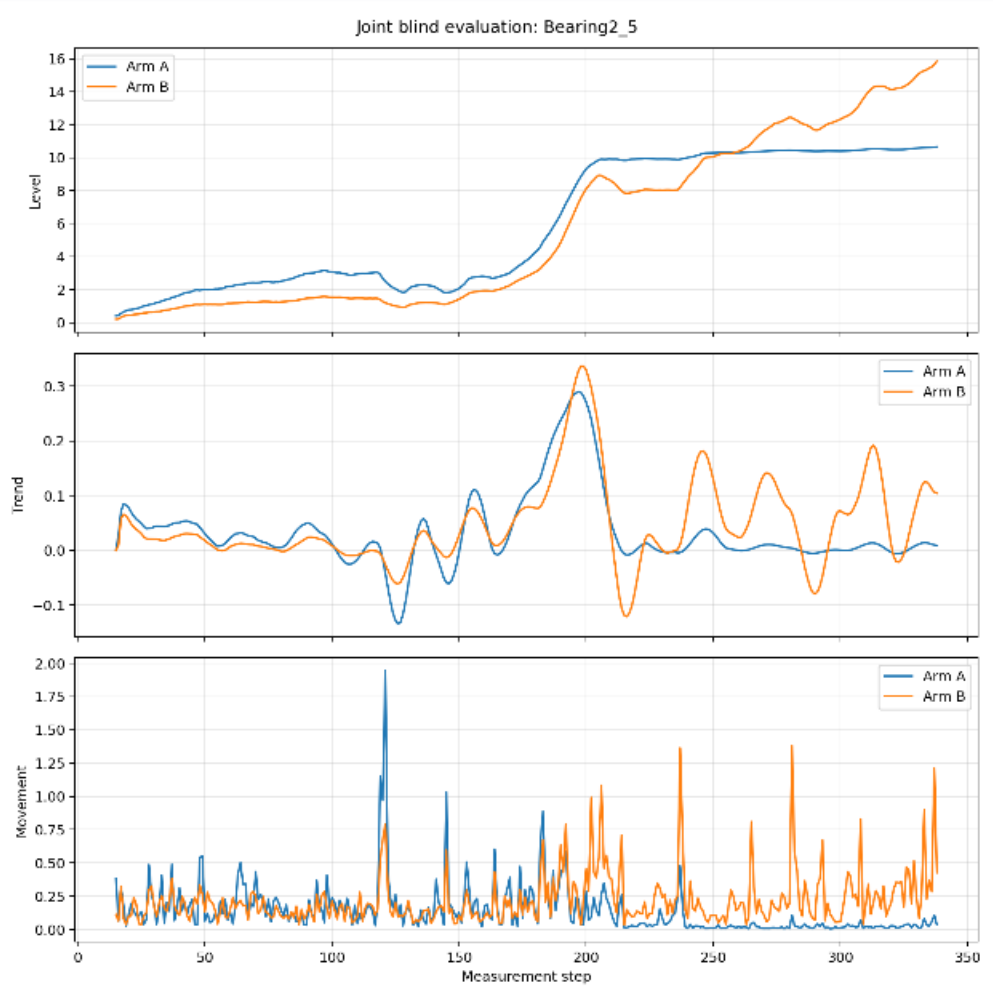
ρ = 0.946
Source normalization + source autoencoder

ARM B · TARGET ADAPTED

ρ = 0.964
Same architecture retrained on target development bearings

Δρ = +0.018

descriptive—not a significance claim



Archived Level, Trend and Movement trajectories for the same blind input sequence.

Source: results/2026-09-12_bearing2_5_blind

A third operating condition exposes domain-shift limits



The figure shows relative Level trajectories on development bearings only; it is not a blind-test result.

FACT

Frozen-model development ρ:
B3_1 = −0.391 · B3_2 = +0.894 · B3_3 = −0.564
The direction is not stable across bearings.

OBSERVATION

Reconstruction-oriented adaptation did not reliably repair the sign instability. More model complexity alone was not sufficient.

NEXT VALIDATION

Evaluate health-aware objectives and domain-aware representations with frozen protocols before reading protected test bearings.

Source: 2026-09-18 Condition 3 frozen development study

TRANSFER TO SAFE PERCEPTION

What this PHM project demonstrates for safe human detection

This is not a computer-vision or human-detection project. The transferable value is the validation discipline.

PHM evidence	Transfer to RGB / depth / point-cloud perception
Reproducible test harness	Repeatable benchmarks across scenes, sensors and model versions
LOBO + cross-condition splits	Subject-, environment- and sensor-level generalization tests
Leakage controls + locked blind runs	Traceable validation evidence for safety-related decisions
Domain-shift failure analysis	Identify when a detector or fusion pipeline leaves its validated domain
Python / PyTorch pipeline	Implement, profile and compare perception-model experiments

Yuxin Song

M.Sc. Autonomous Systems · University of Stuttgart

st196624@stud.uni-stuttgart.de

github.com/yuxinsong2001/state-interpreter